c h a p t e r **8:** Conservation of Energy

8.1 Analysis Model: Nonisolated System (Energy)

8.2 Analysis Model: Isolated System (Energy)

8.3 Situations Involving Kinetic Friction

8.4 Changes in Mechanical Energy for Nonconservative Forces

8.5 Power

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We now consider analyzing physical situations using the energy approach for two types of systems: ***nonisolated and isolated systems.***

- **For nonisolated systems**, we shall investigate ways that energy can cross the boundary of the system, resulting in a change in the system's total energy.
- **In isolated systems**, energy does not cross the boundary of the system. For these systems, the total energy of the system is constant.

8.1 Analysis Model: Nonisolated System (Energy)

If we choose the object as the system, this very simple situation is the first example of a ***nonisolated system, for which energy crosses* the boundary of the system** during some time interval due to an interaction with the environment. This scenario is common in physics problems. **If a system does not interact with its environment, it is an *isolated system***, *which we will study in Section* 8.2.

- **The work–kinetic energy theorem is our first example of an energy equation appropriate for a nonisolated system.**
- In the case of that theorem, the interaction of the system with its environment is the work done by the external force, and the quantity in the system that changes is the kinetic energy.

We illustrate mechanisms to transfer energy in Figure 8.1 and summarize them as follows.

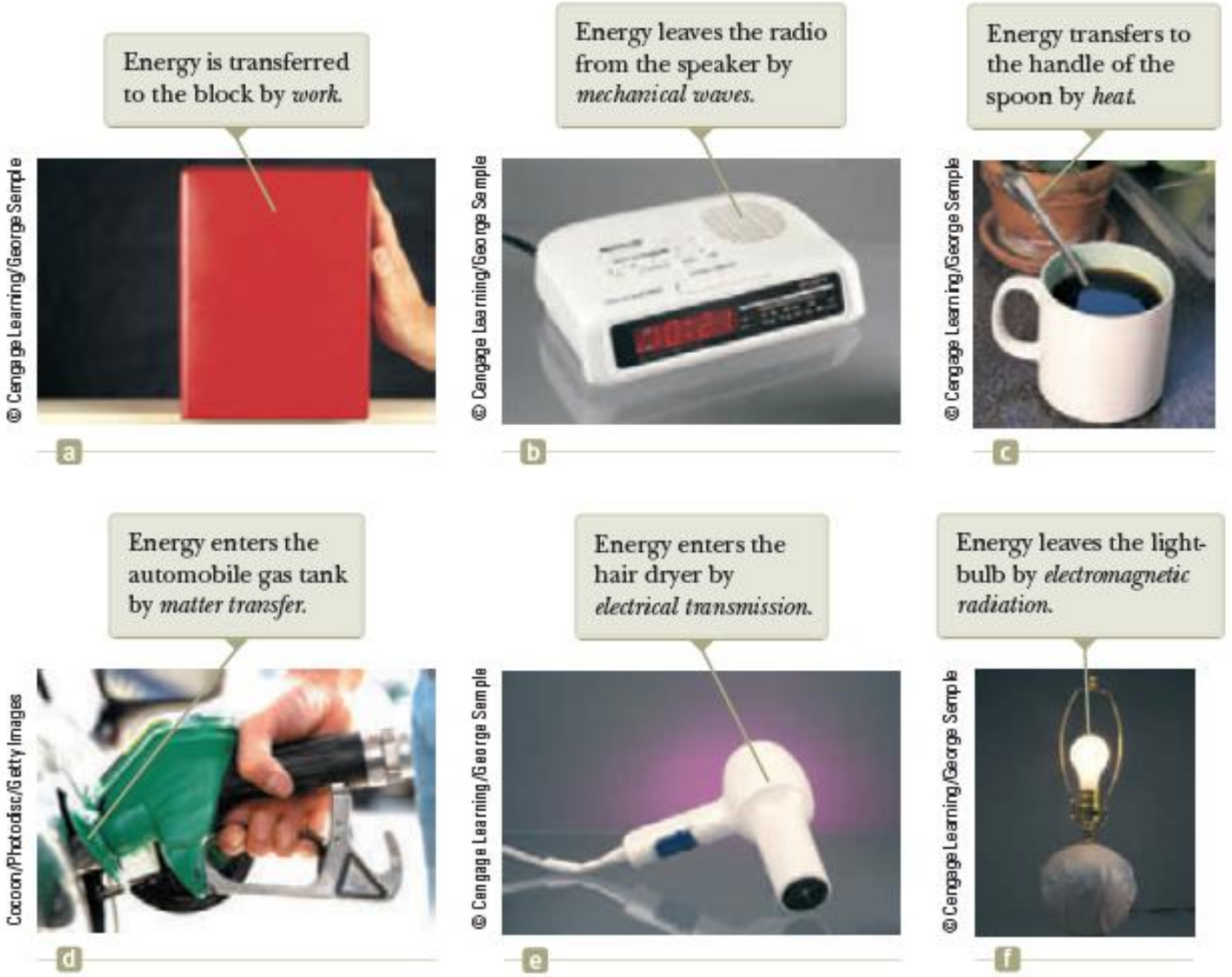


Figure 8.1 Energy transfer mechanisms. In each case, the system into which or from which energy is transferred is indicated.

if the total amount of energy in a system changes, it can *only* be because energy has crossed the boundary of the system by a transfer mechanism such as one of the methods listed above.

- **conservation of energy equation**

$$\Delta E_{system} = \sum T$$

$$\Delta K + \Delta U + \Delta E_{int} = W + Q + T_{MW} + T_{MT} + T_{ET} + T_{ER} \qquad (8.2)$$

which is the primary mathematical representation of the energy version of the analysis model of the **nonisolated system.**

8.2 Analysis Model: Isolated System (Energy)

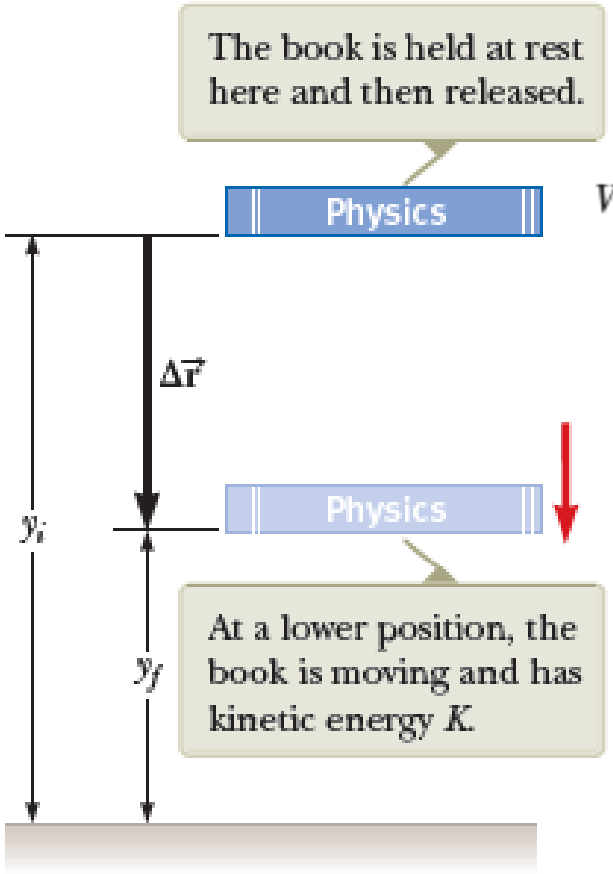


Figure 8.2 A book is released from rest and falls due to work done by the gravitational force on the book.

$$W = \Delta U_g$$

$$W_{\text{on book}} = (m\vec{\mathbf{g}}) \cdot \Delta\vec{\mathbf{r}} = (-mg\hat{\mathbf{j}}) \cdot [(y_f - y_i)\hat{\mathbf{j}}] = mgy_i - mgy_f \qquad (8.3)$$

$$W_{\text{on book}} = \Delta K_{\text{book}}$$

$$\Delta K_{\text{book}} = mgy_i - mgy_f$$

$$mgy_i - mgy_f = -(mgy_f - mgy_i) = -\Delta U_g$$

$$\Delta K = -\Delta U_g$$

$$\Delta K + \Delta U_g = 0$$

$$\Delta E_{\text{mech}} = 0$$

Let us now write the changes in energy in Equation 8.6 explicitly:

$$(K_f - K_i) + (U_f - U_i) = 0$$

$$K_f + U_f = K_i + U_i \tag{8.9}$$

For the gravitational situation of the falling book, Equation 8.9 can be written as

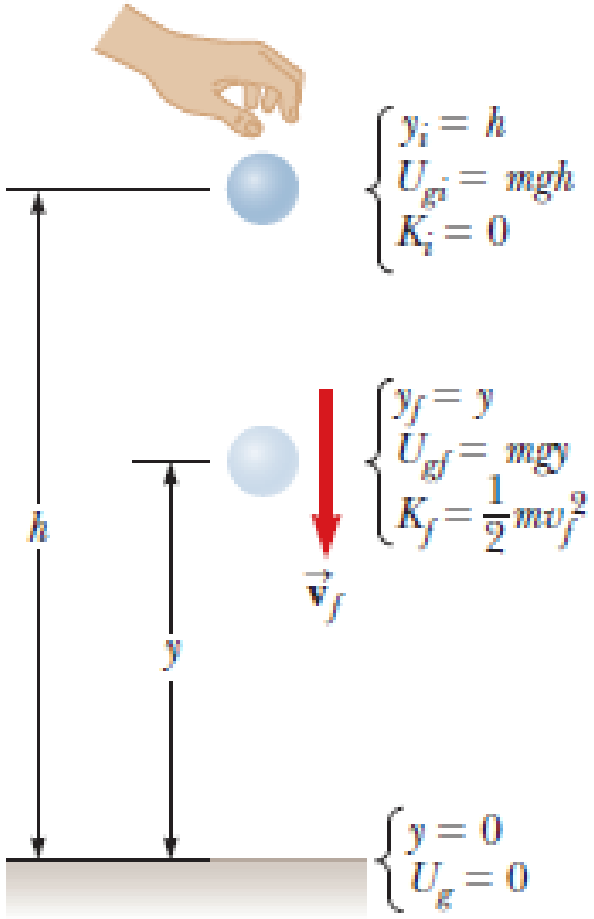
$$\tfrac{1}{2}mv_f^2 + mgy_f = \tfrac{1}{2}mv_i^2 + mgy_i$$

As the book falls to the Earth, the book–Earth system loses potential energy and gains kinetic energy such that the total of the two types of energy always remains constant: $E_{\text{total},i} = E_{\text{total},f}$.

Example 8.1 Ball in Free Fall AM

A ball of mass m is dropped from a height h above the ground as shown in Figure 8.4.

(A) Neglecting air resistance, determine the speed of the ball when it is at a height y above the ground. Choose the system as the ball and the Earth.



$$\Delta K + \Delta U_g = 0$$

$$\left(\tfrac{1}{2}mv_f^2 - 0\right) + (mgy - mgh) = 0$$

$$v_f^2 = 2g(h - y) \quad \rightarrow \quad v_f = \sqrt{2g(h - y)}$$

(B) Find the speed of the ball again at height *y* *by choosing the ball as the system.*

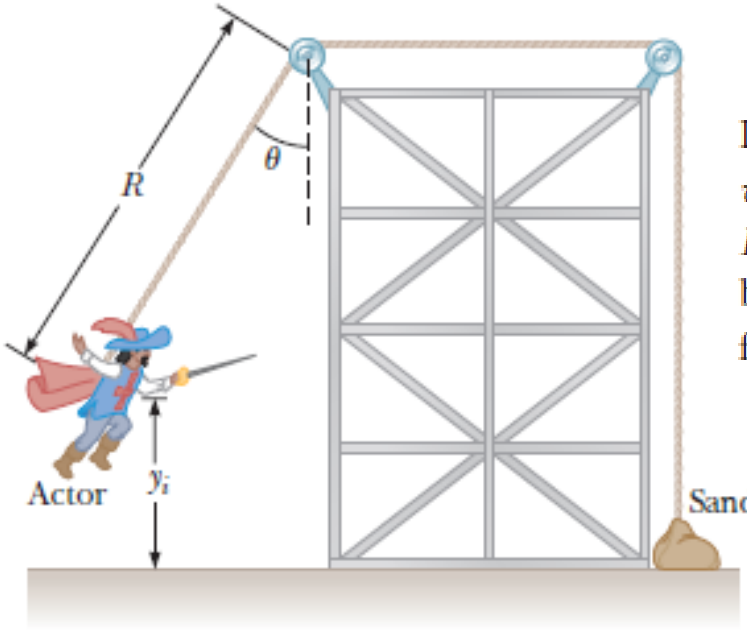
$$\Delta K = W$$

$$\left(\tfrac{1}{2}mv_f^2 - 0\right) = \vec{\mathbf{F}}_g \cdot \Delta\vec{\mathbf{r}} = -mg\hat{\mathbf{j}} \cdot \Delta y\hat{\mathbf{j}}$$
$$= -mg\Delta y = -mg(y - h) = mg(h - y)$$

$$v_f^2 = 2g(h - y) \quad \rightarrow \quad v_f = \sqrt{2g(h - y)}$$

Example 8.2 A Grand Entrance AM

You are designing an apparatus to support an actor of mass 65.0 kg who is to "fly" down to the stage during the performance of a play. You attach the actor's harness to a 130-kg sandbag by means of a lightweight steel cable running smoothly over two frictionless pulleys as in Figure 8.5a. You need 3.00 m of cable between the harness and the nearest pulley so that the pulley can be hidden behind a curtain. For the apparatus to work successfully, the sandbag must never lift above the floor as the actor swings from above the stage to the floor. Let us call the initial angle that the actor's cable makes with the vertical θ. What is the maximum value θ can have before the sandbag lifts off the floor?



a

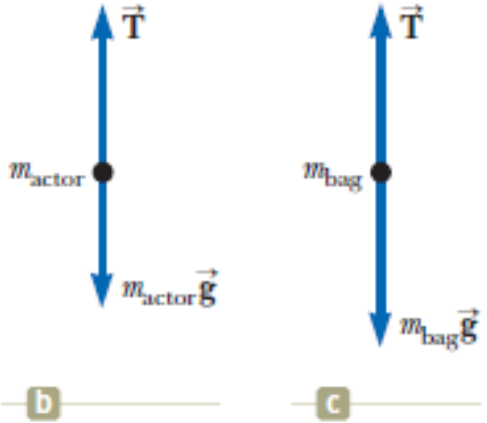
$$\Delta K + \Delta U_g = 0$$

Let y_i be the initial height of the actor above the floor and v_f be his speed at the instant before he lands. (Notice that $K_i = 0$ because the actor starts from rest and that $U_f = 0$ because we define the configuration of the actor at the floor as having a gravitational potential energy of zero.)

$$(1)\quad \left(\tfrac{1}{2}m_{\text{actor}}v_f^2 - 0\right) + \left(0 - m_{\text{actor}}gy_i\right) = 0$$

From the geometry in Figure 8.5a, notice that $y_f = 0$, so $y_i = R - R\cos\theta = R(1 - \cos\theta)$. Use this relationship in Equation (1) and solve for v_f^2:

$$(2)\quad v_f^2 = 2gR(1 - \cos\theta)$$



b c

$$\sum F_y = T - m_{\text{actor}}g = m_{\text{actor}}\frac{v_f^2}{R}$$

$$(3)\quad T = m_{\text{actor}}g + m_{\text{actor}}\frac{v_f^2}{R}$$

$$m_{\text{bag}}g = m_{\text{actor}}g + m_{\text{actor}}\frac{2gR(1 - \cos\theta)}{R}$$

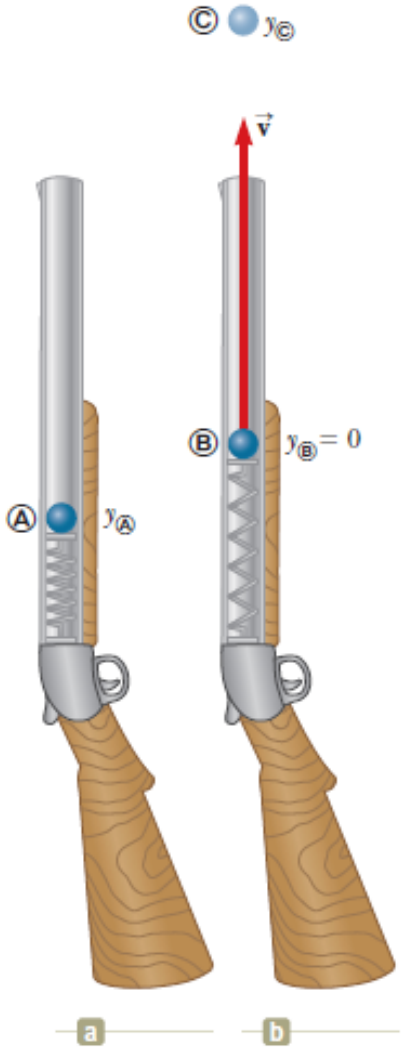
$$\cos\theta = \frac{3m_{\text{actor}} - m_{\text{bag}}}{2m_{\text{actor}}} = \frac{3(65.0\text{ kg}) - 130\text{ kg}}{2(65.0\text{ kg})} = 0.500$$

$$\theta = 60.0^\circ$$

Example 8.3 The Spring-Loaded Popgun AM

The launching mechanism of a popgun consists of a trigger-released spring (Fig. 8.6a). The spring is compressed to a position y_A, and the trigger is fired. The projectile of mass m rises to a position y_C above the position at which it leaves the spring, indicated in Figure 8.6b as position $y_B = 0$. Consider a firing of the gun for which $m = 35.0$ g, $y_A = -0.120$ m, and $y_C = 20.0$ m.

(A) Neglecting all resistive forces, determine the spring constant.



(1) $\Delta K + \Delta U_g + \Delta U_s = 0$

$$(0 - 0) + (mgy_C - mgy_A) + (0 - \tfrac{1}{2}kx^2) = 0$$

$$k = \frac{2mg(y_C - y_A)}{x^2}$$

$$k = \frac{2(0.035\ 0\ \text{kg})(9.80\ \text{m/s}^2)[20.0\ \text{m} - (-0.120\ \text{m})]}{(0.120\ \text{m})^2} = 958\ \text{N/m}$$

(B) Find the speed of the projectile as it moves through the equilibrium position Ⓑ of the spring as shown in Figure 8.6b.

Write Equation (1) again for the system between points Ⓐ and Ⓑ:

$$\Delta K + \Delta U_g + \Delta U_s = 0$$

Substitute for the initial and final energies:

$$\left(\tfrac{1}{2}mv_{Ⓑ}^2 - 0\right) + \left(0 - mgy_{Ⓐ}\right) + \left(0 - \tfrac{1}{2}kx^2\right) = 0$$

Solve for $v_{Ⓑ}$:

$$v_{Ⓑ} = \sqrt{\frac{kx^2}{m} + 2gy_{Ⓐ}}$$

Substitute numerical values:

$$v_{Ⓑ} = \sqrt{\frac{(958\ \text{N/m})(0.120\ \text{m})^2}{(0.035\ 0\ \text{kg})} + 2(9.80\ \text{m/s}^2)(-0.120\ \text{m})} = 19.8\ \text{m/s}$$

8.3 Situations Involving Kinetic Friction

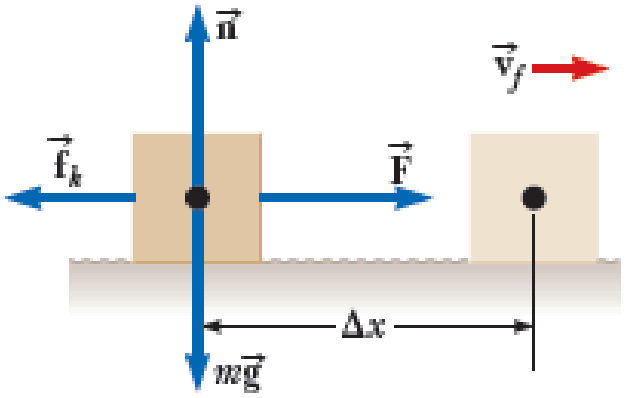
$$\sum W_{\text{other forces}} - f_k d = \Delta K \qquad (8.13)$$

Example 8.4 A Block Pulled on a Rough Surface

AM

A 6.0-kg block initially at rest is pulled to the right along a horizontal surface by a constant horizontal force of 12 N.

(A) Find the speed of the block after it has moved 3.0 m if the surfaces in contact have a coefficient of kinetic friction of 0.15.



a

$$\sum W_{\text{other forces}} = W_F = F\,\Delta x$$

$$\sum F_y = 0 \;\rightarrow\; n - mg = 0 \;\rightarrow\; n = mg$$

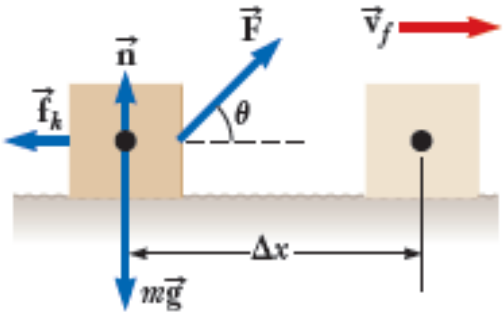
$$f_k = \mu_k n = \mu_k mg = (0.15)(6.0\text{ kg})(9.80\text{ m/s}^2) = 8.82\text{ N}$$

$$F\Delta x = \Delta K + \Delta E_{\text{int}} = \left(\tfrac{1}{2}mv_f^2 - 0\right) + f_k d$$

$$v_f = \sqrt{\frac{2}{m}\left(-f_k d + F\,\Delta x\right)}$$

$$v_f = \sqrt{\frac{2}{6.0\text{ kg}}[-(8.82\text{ N})(3.0\text{ m}) + (12\text{ N})(3.0\text{ m})]} = 1.8\text{ m/s}$$

(B) Suppose the force $\vec{F}$ is applied at an angle θ as shown in Figure 8.8b. At what angle should the force be applied to achieve the largest possible speed after the block has moved 3.0 m to the right?



b

$$(1)\quad \sum W_{\text{other forces}} = W_F = F\Delta x\cos\theta = Fd\cos\theta$$

$$\sum F_y = n + F\sin\theta - mg = 0$$

$$(2)\quad n = mg - F\sin\theta$$

$$W_F = \Delta K + \Delta E_{\text{int}} = (K_f - 0) + f_k d \;\rightarrow\; K_f = W_F - f_k d$$

$$K_f = Fd\cos\theta - \mu_k nd = Fd\cos\theta - \mu_k(mg - F\sin\theta)d$$

$$\frac{dK_f}{d\theta} = -Fd\sin\theta - \mu_k(0 - F\cos\theta)d = 0$$

$$-\sin\theta + \mu_k\cos\theta = 0$$

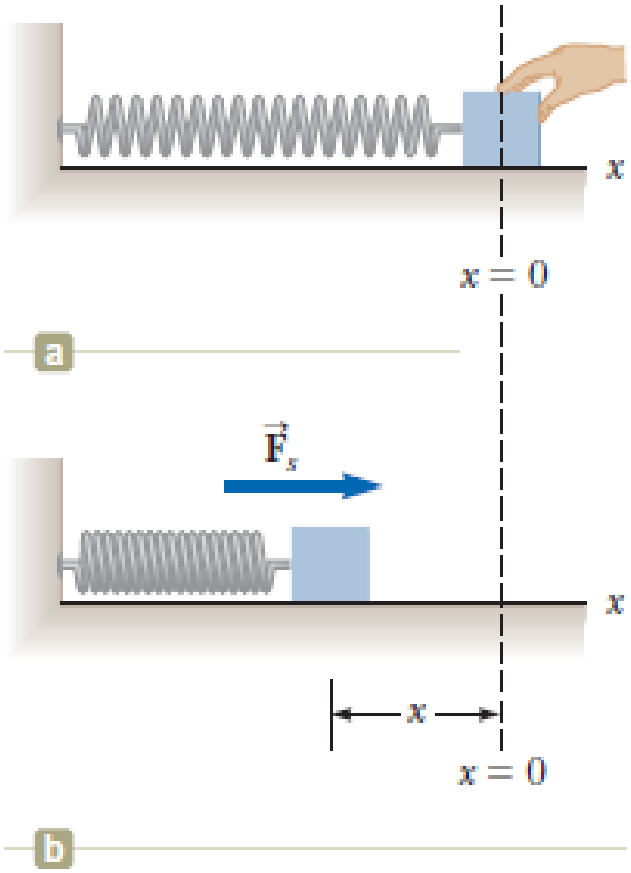
$$\tan\theta = \mu_k$$

$$\theta = \tan^{-1}(\mu_k) = \tan^{-1}(0.15) = 8.5°$$

Example 8.6 A Block–Spring System AM

A block of mass 1.6 kg is attached to a horizontal spring that has a force constant of 1 000 N/m as shown in Figure 8.9a. The spring is compressed 2.0 cm and is then released from rest as in Figure 8.9b.

(A) Calculate the speed of the block as it passes through the equilibrium position $x = 0$ if the surface is frictionless.



$$W_s = \tfrac{1}{2}kx_{\max}^2$$

$$W_s = \tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_i^2$$

$$v_f = \sqrt{v_i^2 + \frac{2}{m}W_s} = \sqrt{v_i^2 + \frac{2}{m}\left(\tfrac{1}{2}kx_{\max}^2\right)}$$

$$v_f = \sqrt{0 + \frac{2}{1.6\ \text{kg}}\left[\tfrac{1}{2}(1\,000\ \text{N/m})(0.020\ \text{m})^2\right]} = \boxed{0.50\ \text{m/s}}$$

(B) Calculate the speed of the block as it passes through the equilibrium position if a constant friction force of 4.0 N retards its motion from the moment it is released.

$$W_s = \Delta K + \Delta E_{\text{int}} = \left(\tfrac{1}{2}mv_f^2 - 0\right) + f_k d$$

$$v_f = \sqrt{\frac{2}{m}\left(W_s - f_k d\right)}$$

$$v_f = \sqrt{\frac{2}{m}\left(\tfrac{1}{2}kx_{\max}^2 - f_k d\right)}$$

$$v_f = \sqrt{\frac{2}{1.6\ \text{kg}}\left[\tfrac{1}{2}(1\,000\ \text{N/m})(0.020\ \text{m})^2 - (4.0\ \text{N})(0.020\ \text{m})\right]} = 0.39\ \text{m/s}$$

8.4 Changes in Mechanical Energy for Nonconservative Forces

In general, if a nonconservative force acts within an isolated system,

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0 \tag{8.16}$$

If the system in which nonconservative forces act is nonisolated and the external influence on the system is by means of work, the generalization of Equation 8.13 is

$$\sum W_{\text{other forces}} - f_k d = \Delta E_{\text{mech}}$$

This equation, with the help of Equations 8.7 and 8.14, can be written as

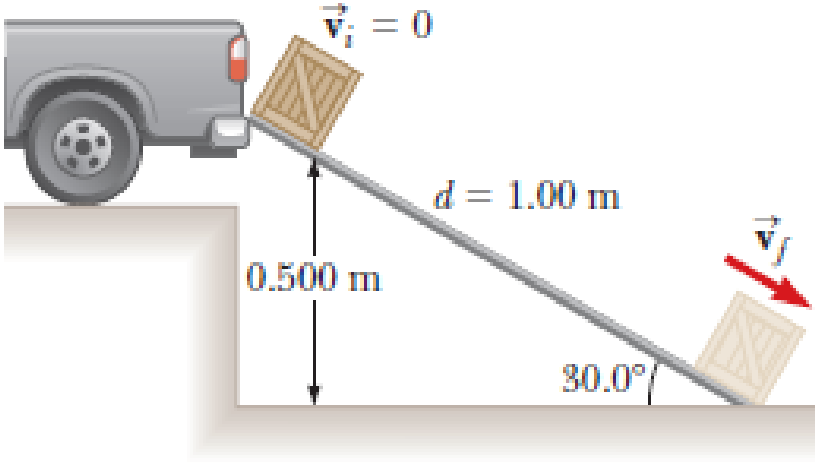
$$\sum W_{\text{other forces}} = W = \Delta K + \Delta U + \Delta E_{\text{int}} \tag{8.17}$$

Example 8.7 Crate Sliding Down a Ramp

AM

A 3.00-kg crate slides down a ramp. The ramp is 1.00 m in length and inclined at an angle of 30.0° as shown in Figure 8.10. The crate starts from rest at the top, experiences a constant friction force of magnitude 5.00 N, and continues to move a short distance on the horizontal floor after it leaves the ramp.

(A) Use energy methods to determine the speed of the crate at the bottom of the ramp.



$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$

$$\left(\tfrac{1}{2}mv_f^2 - 0\right) + \left(0 - mgy_i\right) + f_k d = 0$$

$$(1) \quad v_f = \sqrt{\frac{2}{m}\left(mgy_i - f_k d\right)}$$

$$v_f = \sqrt{\frac{2}{3.00\ \text{kg}}\left[(3.00\ \text{kg})(9.80\ \text{m/s}^2)(0.500\ \text{m}) - (5.00\ \text{N})(1.00\ \text{m})\right]} = 2.54\ \text{m/s}$$

(B) How far does the crate slide on the horizontal floor if it continues to experience a friction force of magnitude 5.00 N?

$$\Delta K + \Delta E_{\text{int}} = 0$$

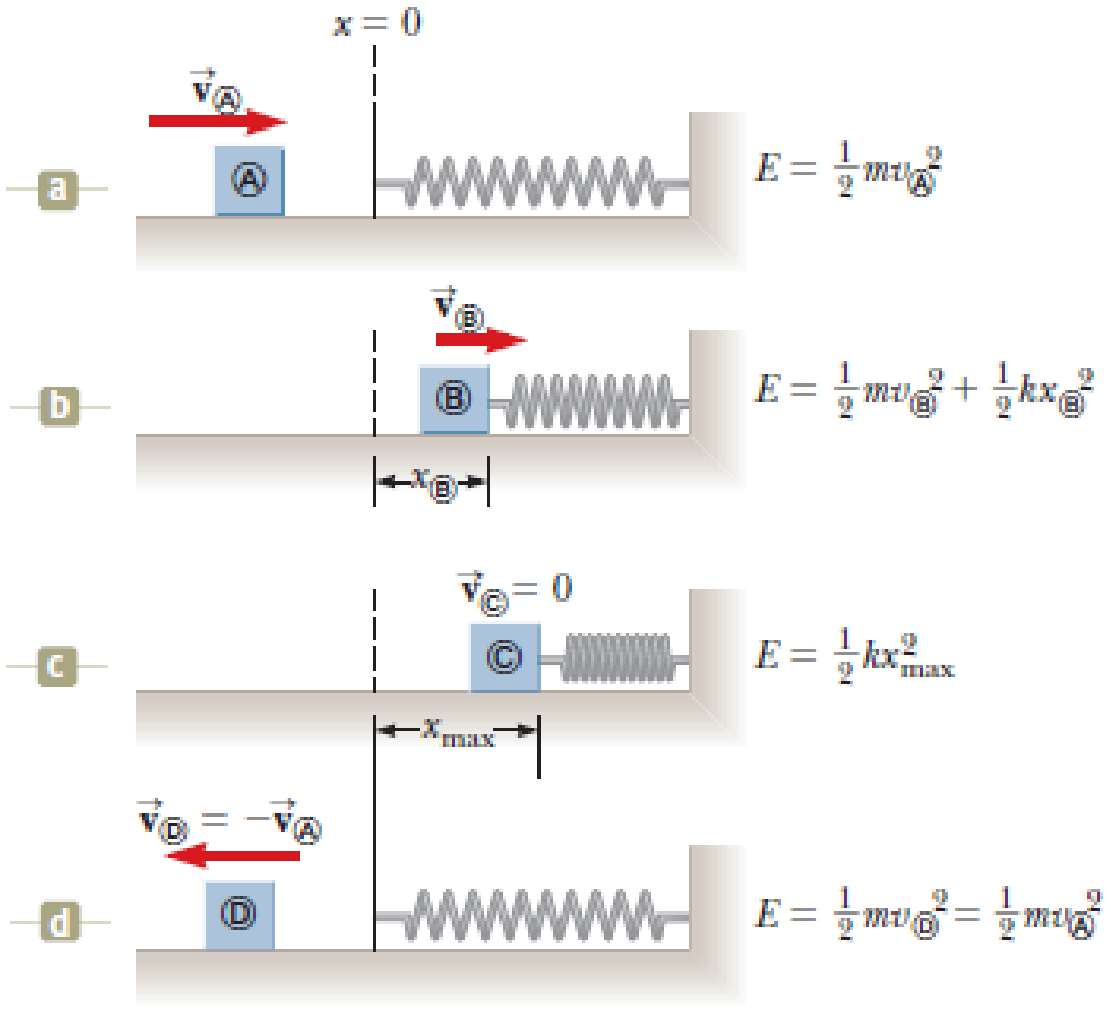
$$\left(0 - \tfrac{1}{2}mv_i^2\right) + f_k d = 0$$

$$d = \frac{mv_i^2}{2f_k} = \frac{(3.00\ \text{kg})(2.54\ \text{m/s})^2}{2(5.00\ \text{N})} = 1.94\ \text{m}$$

Example 8.8 Block–Spring Collision AM

A block having a mass of 0.80 kg is given an initial velocity $v_A = 1.2$ m/s to the right and collides with a spring whose mass is negligible and whose force constant is $k = 50$ N/m as shown in Figure 8.11.

(A) Assuming the surface to be frictionless, calculate the maximum compression of the spring after the collision.



$$\Delta K + \Delta U = 0$$

$$\left(0 - \tfrac{1}{2} m v_{\text{Ⓐ}}^2\right) + \left(\tfrac{1}{2} k x_{\max}^2 - 0\right) = 0$$

$$x_{\max} = \sqrt{\frac{m}{k}}\, v_{\text{Ⓐ}} = \sqrt{\frac{0.80\ \text{kg}}{50\ \text{N/m}}}(1.2\ \text{m/s}) = 0.15\ \text{m}$$

(B) Suppose a constant force of kinetic friction acts between the block and the surface, with $\mu_k = 0.50$. If the speed of the block at the moment it collides with the spring is $v_{\text{Ⓐ}} = 1.2$ m/s, what is the maximum compression $x_{\text{©}}$ in the spring?

$$f_k = \mu_k n = \mu_k mg$$

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0$$

$$\left(0 - \tfrac{1}{2} m v_{\text{Ⓐ}}^2\right) + \left(\tfrac{1}{2} k x_{\text{©}}^2 - 0\right) + \mu_k mg x_{\text{©}} = 0$$

$$k x_{\text{©}}^2 + 2\mu_k mg x_{\text{©}} - m v_{\text{Ⓐ}}^2 = 0$$

$$50 x_{\text{©}}^2 + 2(0.50)(0.80)(9.80) x_{\text{©}} - (0.80)(1.2)^2 = 0$$

$$50 x_{\text{©}}^2 + 7.84 x_{\text{©}} - 1.15 = 0$$

Solving the quadratic equation for $x_{\text{©}}$ gives $x_{\text{©}} = 0.092$ m and $x_{\text{©}} = -0.25$ m. The physically meaningful root is $x_{\text{©}} = 0.092$ m.

Example 8.9 Connected Blocks in Motion AM

Two blocks are connected by a light string that passes over a frictionless pulley as shown in Figure 8.12. The block of mass m_1 lies on a horizontal surface and is connected to a spring of force constant k. The system is released from rest when the spring is unstretched. If the hanging block of mass m_2 falls a distance h before coming to rest, calculate the coefficient of kinetic friction between the block of mass m_1 and the surface.

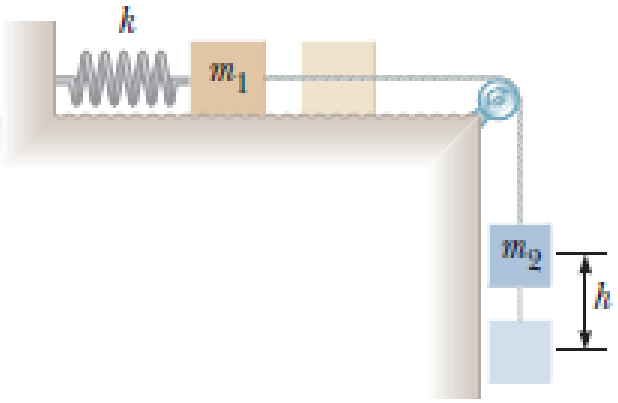


Figure 8.12 (Example 8.9) As the hanging block moves from its highest elevation to its lowest, the system loses gravitational potential energy but gains elastic potential energy in the spring. Some mechanical energy is transformed to internal energy because of friction between the sliding block and the surface.

$$(1)\quad \Delta U_g + \Delta U_s + \Delta E_{\text{int}} = 0$$

$$(0 - m_2gh) + \left(\tfrac{1}{2}kh^2 - 0\right) + f_k h = 0$$

$$-m_2gh + \tfrac{1}{2}kh^2 + \mu_k m_1 gh = 0$$

$$\mu_k = \frac{m_2g - \frac{1}{2}kh}{m_1 g}$$

8.5 Power

The time rate of energy transfer is called the **instantaneous power** P and is defined as

$$P \equiv \frac{dE}{dt} \tag{8.18}$$

We will focus on work as the energy transfer method in this discussion, but keep in mind that the notion of power is valid for *any* means of energy transfer discussed in Section 8.1. If an external force is applied to an object (which we model as a particle) and if the work done by this force on the object in the time interval Δt is W, the **average power** during this interval is

$$P_{avg} = \frac{W}{\Delta t}$$

Therefore, in Conceptual Example 7.7, although the same work is done in rolling the refrigerator up both ramps, less power is required for the longer ramp.

In a manner similar to the way we approached the definition of velocity and acceleration, the instantaneous power is the limiting value of the average power as Δt approaches zero:

$$P = \lim_{\Delta t \to 0} \frac{W}{\Delta t} = \frac{dW}{dt}$$

where we have represented the infinitesimal value of the work done by dW. We find from Equation 7.3 that $dW = \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$. Therefore, the instantaneous power can be written

$$P = \frac{dW}{dt} = \vec{\mathbf{F}} \cdot \frac{d\vec{\mathbf{r}}}{dt} = \vec{\mathbf{F}} \cdot \vec{\mathbf{v}} \tag{8.19}$$

where $\vec{\mathbf{v}} = d\vec{\mathbf{r}}/dt$.

The SI unit of power is joules per second (J/s), also called the **watt** (W) after James Watt:

$$1\ \text{W} = 1\ \text{J/s} = 1\ \text{kg} \cdot \text{m}^2/\text{s}^3$$

A unit of power in the U.S. customary system is the **horsepower** (hp):

$$1\ \text{hp} = 746\ \text{W}$$

A unit of energy (or work) can now be defined in terms of the unit of power. One **kilowatt-hour** (kWh) is the energy transferred in 1 h at the constant rate of 1 kW = 1 000 J/s. The amount of energy represented by 1 kWh is

$$1\ \text{kWh} = (10^3\ \text{W})(3\,600\ \text{s}) = 3.60 \times 10^6\ \text{J}$$

Example 8.11 Power Delivered by an Elevator Motor

AM

An elevator car (Fig. 8.14a) has a mass of 1 600 kg and is carrying passengers having a combined mass of 200 kg. A constant friction force of 4 000 N retards its motion.

(A) How much power must a motor deliver to lift the elevator car and its passengers at a constant speed of 3.00 m/s?



a



b

$$\sum F_y = T - f - Mg = 0$$

$$T = Mg + f$$

$$P = \vec{\mathbf{T}} \cdot \vec{\mathbf{v}} = Tv = (Mg + f)v$$

$$P = [(1\,800\text{ kg})(9.80\text{ m/s}^2) + (4\,000\text{ N})](3.00\text{ m/s}) = 6.49 \times 10^4\text{ W}$$

(B) What power must the motor deliver at the instant the speed of the elevator is v if the motor is designed to provide the elevator car with an upward acceleration of 1.00 m/s²?

$$\sum F_y = T - f - Mg = Ma$$

$$T = M(a + g) + f$$

$$P = Tv = [M(a + g) + f]v$$

$$P = [(1\,800\ \text{kg})(1.00\ \text{m/s}^2 + 9.80\ \text{m/s}^2) + 4\,000\ \text{N}]v$$

$$= (2.34 \times 10^4)v$$

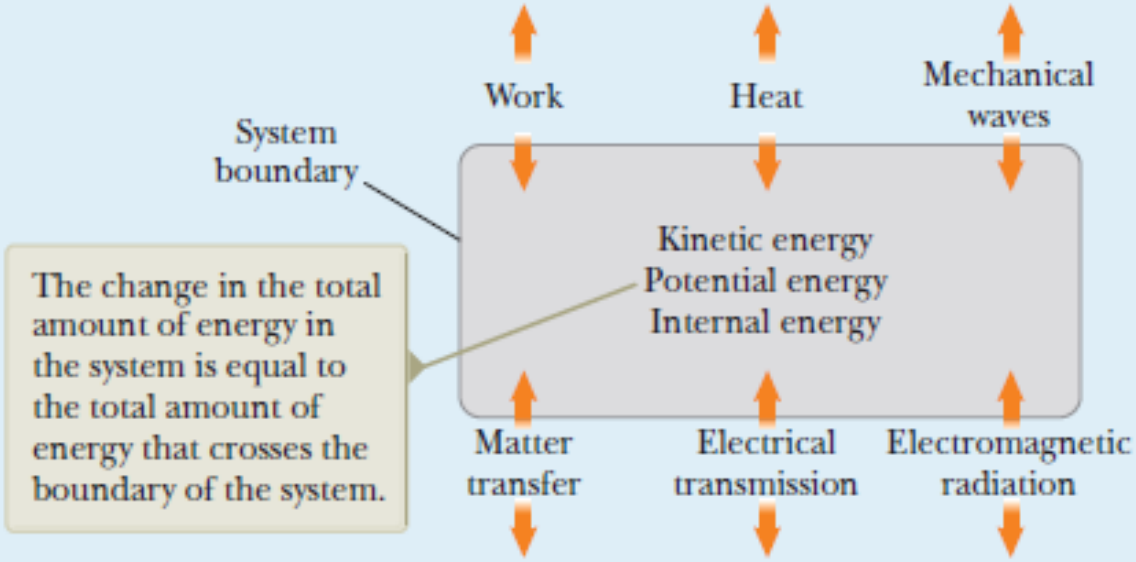
Finalize To compare with part (A), let $v = 3.00$ m/s, giving a power of

$$P = (2.34 \times 10^4\ \text{N})(3.00\ \text{m/s}) = 7.02 \times 10^4\ \text{W}$$

which is larger than the power found in part (A), as expected.

A **nonisolated system** is one for which energy crosses the boundary of the system. An **isolated system** is one for which no energy crosses the boundary of the system.

For a nonisolated system, we can equate the change in the total energy stored in the system to the sum of all the transfers of energy across the system boundary, which is a statement of **conservation of energy.** For an isolated system, the total energy is constant.



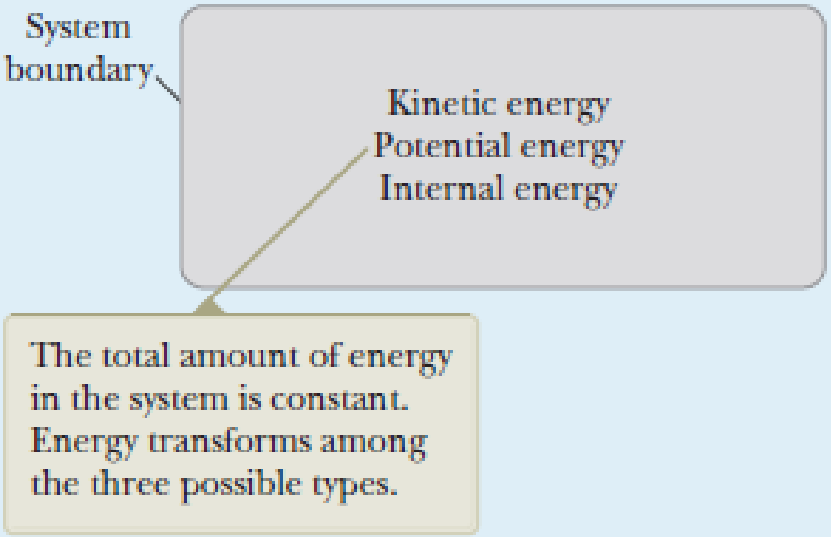
Nonisolated System (Energy). The most general statement describing the behavior of a nonisolated system is the **conservation of energy equation:**

$$\Delta E_{\text{system}} = \Sigma\, T \tag{8.1}$$

Including the types of energy storage and energy transfer that we have discussed gives

$$\Delta K + \Delta U + \Delta E_{\text{int}} = W + Q + T_{\text{MW}} + T_{\text{MT}} + T_{\text{ET}} + T_{\text{ER}} \tag{8.2}$$

For a specific problem, this equation is generally reduced to a smaller number of terms by eliminating the terms that are not appropriate to the situation.



Isolated System (Energy). The total energy of an isolated system is conserved, so

$$\Delta E_{\text{system}} = 0 \qquad \textbf{(8.10)}$$

which can be written as

$$\Delta K + \Delta U + \Delta E_{\text{int}} = 0 \qquad \textbf{(8.16)}$$

If no nonconservative forces act within the isolated system, the mechanical energy of the system is conserved, so

$$\Delta E_{\text{mech}} = 0 \qquad \textbf{(8.8)}$$

which can be written as

$$\Delta K + \Delta U = 0 \qquad \textbf{(8.6)}$$

If a friction force of magnitude f_k acts over a distance d within a system, the change in internal energy of the system is

$$\Delta E_{\text{int}} = f_k d \qquad \textbf{(8.14)}$$

The **instantaneous power** P is defined as the time rate of energy transfer:

$$P \equiv \frac{dE}{dt} \qquad \textbf{(8.18)}$$